

Lecture 35-36: Proofs

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Outline

- Proofs: Formal versus Informal
- Some terminology about proofs
- Proof methods
- Proof of equivalence
- Mistakes in proofs
- Proof strategies
 - Forward and backward reasoning
 - Adapting existing proofs
 - Illustration with tilings
- The role of open problems

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- **Proofs: Formal versus Informal**
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Proofs: Formal Versus Informal

Proofs

A sequence of statements such that

each statement is

either a premise or follows from

the truth of its preceding statements

Proofs: Formal Versus Informal

Peano Arithmetic $(\mathbb{N}, S, +, \cdot)$

$$\forall x \neg Sx = 0 \quad \forall x \forall y Sx = Sy \rightarrow x = y$$

$$\forall x x + 0 = x \quad \forall x \forall y x + Sy = S(x + y)$$

$$\forall x x \cdot 0 = 0 \quad \forall x \forall y x \cdot Sy = x \cdot y + x$$

Induction axiom $(\varphi(0) \wedge \forall x (\varphi(x) \rightarrow \varphi(Sx))) \rightarrow \forall x \varphi(x)$

Problem: Prove $\forall x (0 + x = x)$

Proofs: Formal Versus Informal

Formal proofs: All steps supplied, **long and hard to read for humans**, but **easier for computers** (automated reasoning systems) **to produce**

1. $(\varphi(0) \wedge \forall x (\varphi(x) \rightarrow \varphi(Sx))) \rightarrow \forall x \varphi(x)$ Induction axiom $\varphi(x): 0 + x = x$
2. $\forall x x + 0 = x$ Axiom
3. $0 + 0 = 0$ Universal instantiation of 2
4. $\forall x \forall y x + Sy = S(x + y)$ Axiom
5. $0 + Sc = S(0 + c)$ Universal instantiation of 4
6. $0 + c = c \rightarrow S(0 + c) = Sc$ S is a function
7. $0 + c = c \rightarrow 0 + Sc = Sc$ Transitivity of $=$, 5 and 6
8. $\forall x (0 + x = x \rightarrow 0 + Sx = Sx)$ Universal generalization of 7
9. $0 + 0 = 0 \wedge \forall x (0 + x = x \rightarrow 0 + Sx = Sx)$ Conjunction of 2 and 8
10. $\forall x (0 + x = x)$ Hypothetical syllogism of 9 and 1

Proofs: Formal Versus Informal

Informal proofs: Some steps skipped or combined, **short and easier to explain to humans**, but **hard for computers to produce**

~~1. $(\varphi(0) \wedge \forall x (\varphi(x) \rightarrow \varphi(Sx))) \rightarrow \forall x \varphi(x)$~~

~~Induction axiom~~

~~2. $\forall x x + 0 = x$~~

~~Axiom~~

$$0 + 0 = 0$$

~~Universal instantiation of 2~~

~~4. $\forall x \forall y x + Sy = S(x + y)$~~

~~Axiom~~

~~5. $0 + Sc = S(0 + c)$~~

~~Universal instantiation of 4~~

~~6. $0 + c = c \rightarrow 0 + Sc = S(0 + c) = Sc$~~

 ~~S is a function~~

~~7. $0 + c = c \rightarrow 0 + Sc = Sc$~~

~~Transitivity of $=$, 5 and 6~~

$$\forall x (0 + x = x \rightarrow 0 + Sx = Sx)$$

~~Universal generalization of 7~~

~~9. $0 + 0 = 0 \wedge \forall x (0 + x = x \rightarrow 0 + Sx = Sx)$~~

~~Conjunction of 2 and 8~~

$$\forall x (0 + x = x)$$

~~Hypothetical syllogism of 9 and 1~~

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Some Terminology About Proofs

Theorems Statements that can be **proved to be true**

Propositions **Less important** theorems

Axioms

Statements we **always assume to be true** (e.g. those in Peano arithmetic)

Lemmas

Less important theorems **helpful** in proving the **other results**

Corollaries

Theorems inferred **directly** from some **other** theorems

Conjectures

Statements **believed to be true**, usually with partial evidences or the intuition of an expert, but **not proved yet**

Definitions Define predicates, functions, or relations used in theorems

Some Terminology About Proofs

Universal quantifiers are usually omitted
when theorems are stated

If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$.



For all positive real numbers x and y , if $x > y$, then $x^2 > y^2$.

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Direct Proof

Theorem. $\forall x (\varphi_1(x) \rightarrow \varphi_2(x))$.

Proof. Show that $\varphi_1(c)$ implies $\varphi_2(c)$.

Definition. An integer n is even (odd) if there exists integer k such that $n = 2k$ ($n = 2k+1$).

Theorem. If n is odd, then n^2 is odd.

Proof.

Because n is odd, there exists k such that $n = 2k + 1$.

Then $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(k^2 + 2k) + 1$.

Therefore, n^2 is odd.

Proof by Contraposition

Theorem. $\forall x (\varphi_1(x) \rightarrow \varphi_2(x)).$

Proof. Show that $\neg\varphi_2(c)$ implies $\neg\varphi_1(c)$.

Theorem. Let n be an integer. If $3n + 2$ is odd, then n is odd.

Proof. Suppose n is even. Then $n = 2k$ for some integer k .
Thus, $3n + 2 = 3(2k) + 2 = 2(3k + 1)$.
Therefore, $3n + 2$ is even.

Proof by Contraposition

Theorem. $\forall x (\varphi_1(x) \rightarrow \varphi_2(x)).$

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Therefore, $3n + 2$ is even.

Theorem. Let n be an integer. If n^2 is odd, then n is odd.

Proof. Suppose n is even. Then $n = 2k$ for some integer k .
Thus $n^2 = (2k)^2 = 2(2k^2)$ is even.

Proof by Contradiction

Theorem. $\forall x (\varphi_1(x) \rightarrow \varphi_2(x))$.

Proof. Show that $\varphi_1(c) \wedge \neg \varphi_2(c)$ implies $\psi \wedge \neg \psi$ for some formula ψ .

Definition. A real number r is rational if there are integers m and n such that $n \neq 0$ and $r = m/n$

Theorem. Show that $\sqrt{2}$ is irrational.

Proof.

Proof by Cases

Theorem. $\forall x (\varphi_1(x) \rightarrow \varphi_2(x))$.

Proof. If $\varphi_1 \equiv \varphi_{1,1} \vee \cdots \vee \varphi_{1,n}$, then show $\varphi_{1,1}(c)$ implies $\varphi_2(c)$, $\cdots$, and $\varphi_{1,n}(c)$ implies $\varphi_2(c)$

Theorem. Let n be an integer. Prove that $n^2 \geq n$.

Proof. Distinguish between three cases $n \geq 1$, $n = 0$, $n < 0$.

- 1) Case $n \geq 1$: It follows $n \cdot n \geq 1 \cdot n$, namely, $n^2 \geq n$.
- 2) Case $n = 0$: $n^2 = 0 = n$. Thus $n^2 \geq n$.
- 3) Case $n < 0$: It follows $n^2 > 0 > n$, thus $n^2 \geq n$.

Proof by Cases

Theorem. Prove that the last digit of the square of every integer is 0, 1, 4, 5, 6, or 9.

Proof.

Proof by Cases

Common errors: **Some cases are missing** in the proof

Show that if x is a real number, then x^2 is positive.

Proof.

- 1) Case $x > 0$: $x^2 > 0$.
- 2) Case $x < 0$: $x^2 = (-x) \cdot (-x) > 0$.

Proof by Cases

Common errors: Some cases are missing in the proof

Show that if x is a real number, then x^2 is positive.

Proof.

- 1) Case $x > 0$: $x^2 > 0$.
- 2) Case $x < 0$: $x^2 = (-x) \cdot (-x) > 0$.
- 3) Case $x = 0$: $x^2 = 0$

Proof by Cases

Common errors: Some cases are missing in the proof

Show that if x is a real number, then x^2 is positive.

Proof.

- 1) Case $x > 0$: $x^2 > 0$.
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- 3) Case $x = 0$: $x^2 = 0$

Counterexample



Existence Proofs

Theorem. $\exists x \varphi_1(x)$.

Constructive proof: Show that $\varphi_1(c)$ holds for some c

Theorem. There is a positive integer that can be written as the sum of cubes of positive integers in two different ways.

$$\exists x > 0 \exists y_1 > 0 \exists y_2 > 0 \left(\begin{array}{l} \neg(y_1 = z_1 \wedge y_2 = z_2) \wedge \neg(y_1 = z_2 \wedge y_2 = z_1) \wedge \\ \exists z_1 > 0 \exists z_2 > 0 \quad x = (y_1)^3 + (y_2)^3 \wedge x = (z_1)^3 + (z_2)^3 \end{array} \right)$$

Proof. $1729 = 10^3 + 9^3 = 12^3 + 1^3$

Existence Proofs

Theorem. $\exists x \varphi_1(x)$.

Nonconstructive proof: Show that $\exists x \varphi_1(x)$ holds, without proving $\varphi_1(c)$ holds for a specific c .

Theorem. There are irrationals x and y such that x^y is a rational.

Proof.

Existence Proofs

Theorem. $\exists x \varphi_1(x)$.

Nonconstructive proof: Show that $\exists x \varphi_1(x)$ holds, without proving $\varphi_1(c)$ holds for a specific c .

Theorem. There are irrationals x and y such that x^y is a rational.

Proof.

Question: Is $\sqrt{2}^{\sqrt{2}}$ irrational ? Hilbert's seventh problem (1900), Proved (1934).

Uniqueness Proofs

Theorem. $\exists! x \varphi_1(x)$.

Proof. *Existence.* Show that $\varphi_1(c)$ holds for some c .

Uniqueness. For every c' , if $\varphi_1(c')$, then $c' = c$.

Theorem. If a and b are real numbers with $a \neq 0$, then there is a unique real number r such that $a r + b = 0$.

Proof.

Let $r = -b/a$. Then $a r + b = a (-b/a) + b = -b + b = 0$.

Suppose $a s + b = 0 = a r + b$.

Then $a s = a r$. From $a \neq 0$, it follows $s = r$.

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Proof of Equivalence

Theorem. $\forall x (\varphi_1(x) \leftrightarrow \varphi_2(x))$.

Proof. Show $\forall x (\varphi_1(x) \rightarrow \varphi_2(x))$ and $\forall x (\varphi_2(x) \rightarrow \varphi_1(x))$.

Theorem. Let n be an integer. Then n^2 is odd iff n is odd.

Proof. Show that

- 1) if n is odd, then n^2 is odd, and
- 2) if n^2 is odd then n is odd.

Proof of Equivalence

Theorem. $\forall x \varphi_1(x) \leftrightarrow \varphi_2(x) \cdots \leftrightarrow \varphi_n(x)$.

Proof. Show $\forall x \varphi_1(x) \rightarrow \varphi_2(x)$, $\cdots$, and $\forall x \varphi_n(x) \rightarrow \varphi_1(x)$.

Theorem. $p_1 \leftrightarrow p_2 \leftrightarrow p_3$, where p_1 : n is even, p_2 : $3n+2$ is even, p_3 : n^2 is even.

Proof. Show that

- 1) $p_1 \rightarrow p_2$: From $n = 2k$, it follows that $3n+2 = 2(3k+1)$ is even.
- 2) $p_2 \rightarrow p_3$: From $3n+2 = 2k$, it follows that $3n = 2(k-1)$.
Therefore, $9n^2 = 4(k-1)^2$. So $4 \mid (9n^2)$. It follows n^2 is even.
- 3) $p_3 \rightarrow p_1$: Equivalently, we show $\neg p_1 \rightarrow \neg p_3$.
Suppose n is odd, then $n = 2k+1$.
So $n^2 = (2k+1)^2 = 2(2k^2+2k)+1$ is odd.

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Mistakes in Proofs

A proof for the following statement ?

Let n be an integer. If n^2 is positive, then n is positive.

n^2 is positive.

If n is positive, then n^2 is positive

Therefore, n is positive.

Mistakes in Proofs

A proof for the following statement ?

Let n be an integer. If n^2 is positive, then n is positive.

n^2 is positive. q

If n is positive, then n^2 is positive $p \rightarrow q$

Therefore, n is positive. p

$\frac{\varphi_2}{\varphi_1 \rightarrow \varphi_2}$ is not an inference rule
 $\therefore \varphi_1$

Mistakes in Proofs

A proof for the following statement ?

Let n be an integer. If n is not positive, then n^2 is not positive.

n is not positive.

If n is positive, then n^2 is positive

Therefore, n^2 is not positive.

Mistakes in Proofs

A proof for the following statement ?

Let n be an integer. If n is not positive, then n^2 is not positive.

n is not positive. $\neg p$

If n is positive, then n^2 is positive $p \rightarrow q$

Therefore, n^2 is not positive. $\neg q$

$$\frac{\neg \varphi_1 \quad \varphi_1 \rightarrow \varphi_2}{\therefore \neg \varphi_2}$$
 is not an inference rule

Mistakes in Proofs

Circular reasoning

问：《圣经》是否是可信的？

答：是。因为那是上帝的语言被人记录下来的。

Mistakes in Proofs

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问：《圣经》是否是可信的？

答：是。因为那是上帝的语言被人记录下来的。

问：上帝是否存在？

答：《圣经》中说上帝是存在的，所以上帝是存在的。

Mistakes in Proofs

Circular reasoning

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$q?$

$p \rightarrow q$

问：上帝是否存在？

$p?$

答：《圣经》中说上帝是存在的，所以上帝是存在的。

$q \rightarrow p$

p : 上帝存在

q : 《圣经》可信

Mistakes in Proofs

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问：上帝是否存在？

$p?$

答：《圣经》中说上帝是存在的，所以上帝是存在的。

$q \rightarrow p$

p : 上帝存在

q : 《圣经》可信

What's going wrong ?

Mistakes in Proofs

Circular reasoning

问：《圣经》是否是可信的？

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答：《圣经》中说上帝是存在的，所以上帝是存在的。

p : 上帝存在

q : 《圣经》可信

$q?$

$p \rightarrow q$

$p?$

$q \rightarrow p$

What's going wrong ?

Only show $p \leftrightarrow q$.

It is possible that both p and q are false.

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Forward and Backward Reasoning

Forward reasoning.

Starting from premises, build a sequence of proof steps until reaching the conclusion.

Backward reasoning.

Starting from the conclusion, build a sequence of proof steps backwards, until reaching the premises or the axioms or the already proved theorems.

Forward and Backward Reasoning

Backward reasoning: Examples

Theorem. If x and y are two distinct positive real numbers, then

$$(x + y)/2 > \sqrt{xy}$$

Proof.

$$(x + y)/2 > \sqrt{xy}$$

$$(x + y) > 2\sqrt{xy}$$

$$(x + y)^2 > 4xy$$

$$x^2 + 2xy + y^2 > 4xy$$

$$x^2 - 2xy + y^2 > 0$$

$$(x - y)^2 > 0$$

$$x \neq y$$

Forward and Backward Reasoning

Backward reasoning: Examples

Nim Game

Player Alice and Bob
take turns



Each time a player
removes 1, 2, or 3 stones

The player who removes the last stone wins the game.

A player has a winning strategy if
the player can win no matter what the other player does

Forward and Backward Reasoning

Backward reasoning: Examples

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The player who moves second has a winning strategy.

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Backward reasoning: Examples

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Forward and Backward Reasoning

Backward reasoning: Examples

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Player Alice and Bob
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The player who removes the last stone wins the game.

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The player who moves second has a winning strategy.

Forward and Backward Reasoning

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Player Alice and Bob
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Each time a player
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The player who removes the last stone wins the game.

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The second player has a winning strategy.

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The player who removes the last stone wins the game.

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The second player has a winning strategy.

Forward and Backward Reasoning

Backward reasoning: Examples

Theorem. In a Nim game with 15 stones, the first player (the player who moves first) has a winning strategy.

Proof. Suppose the first player wins.

1st Player
wins

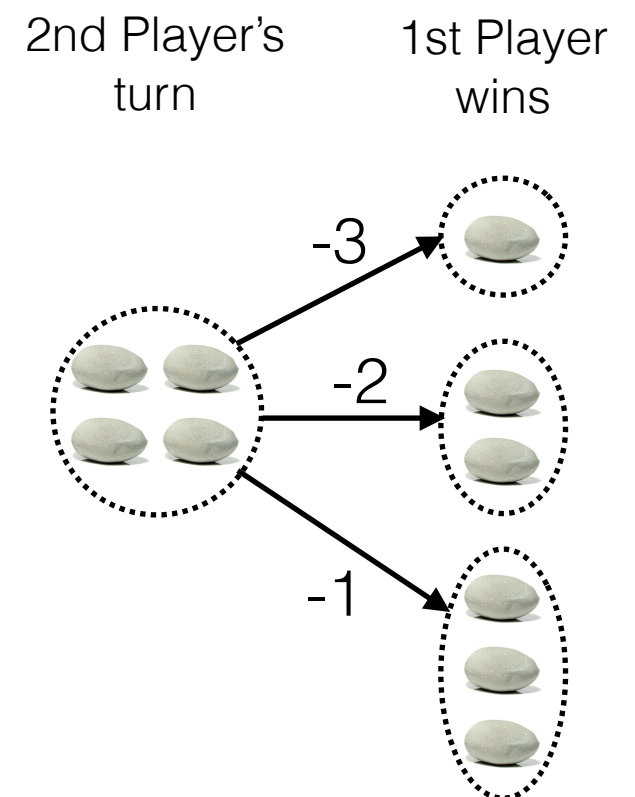


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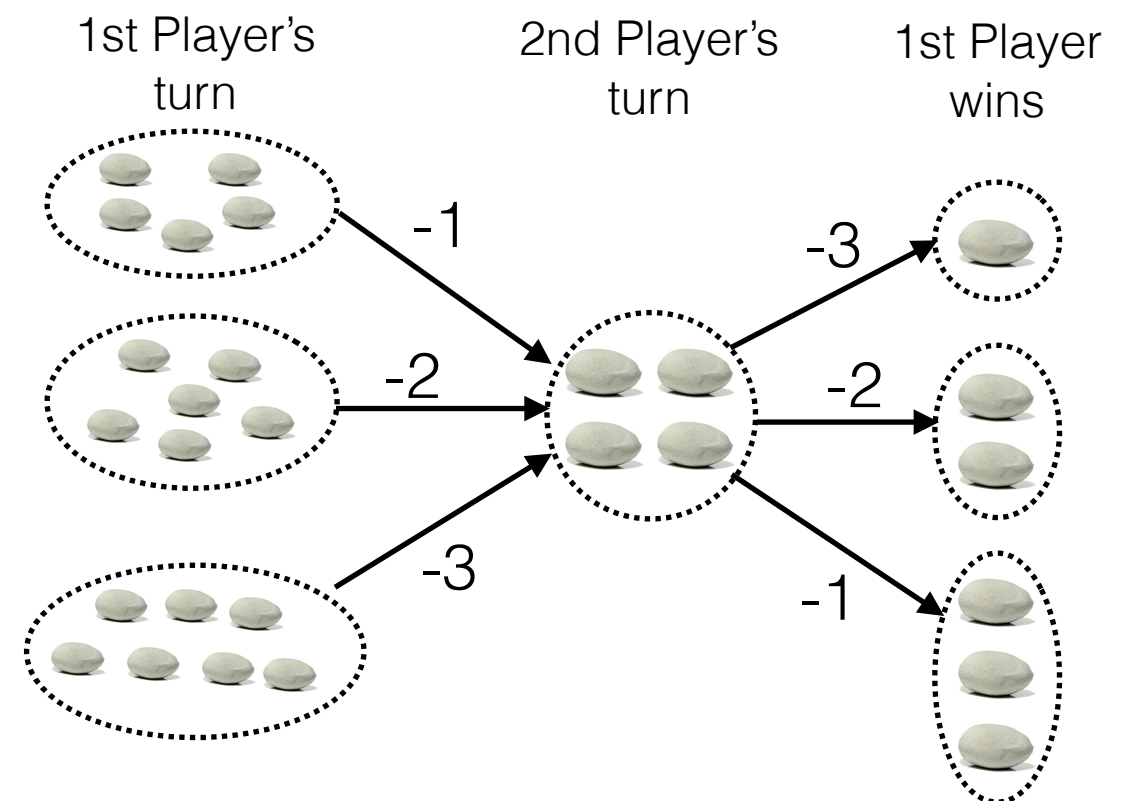


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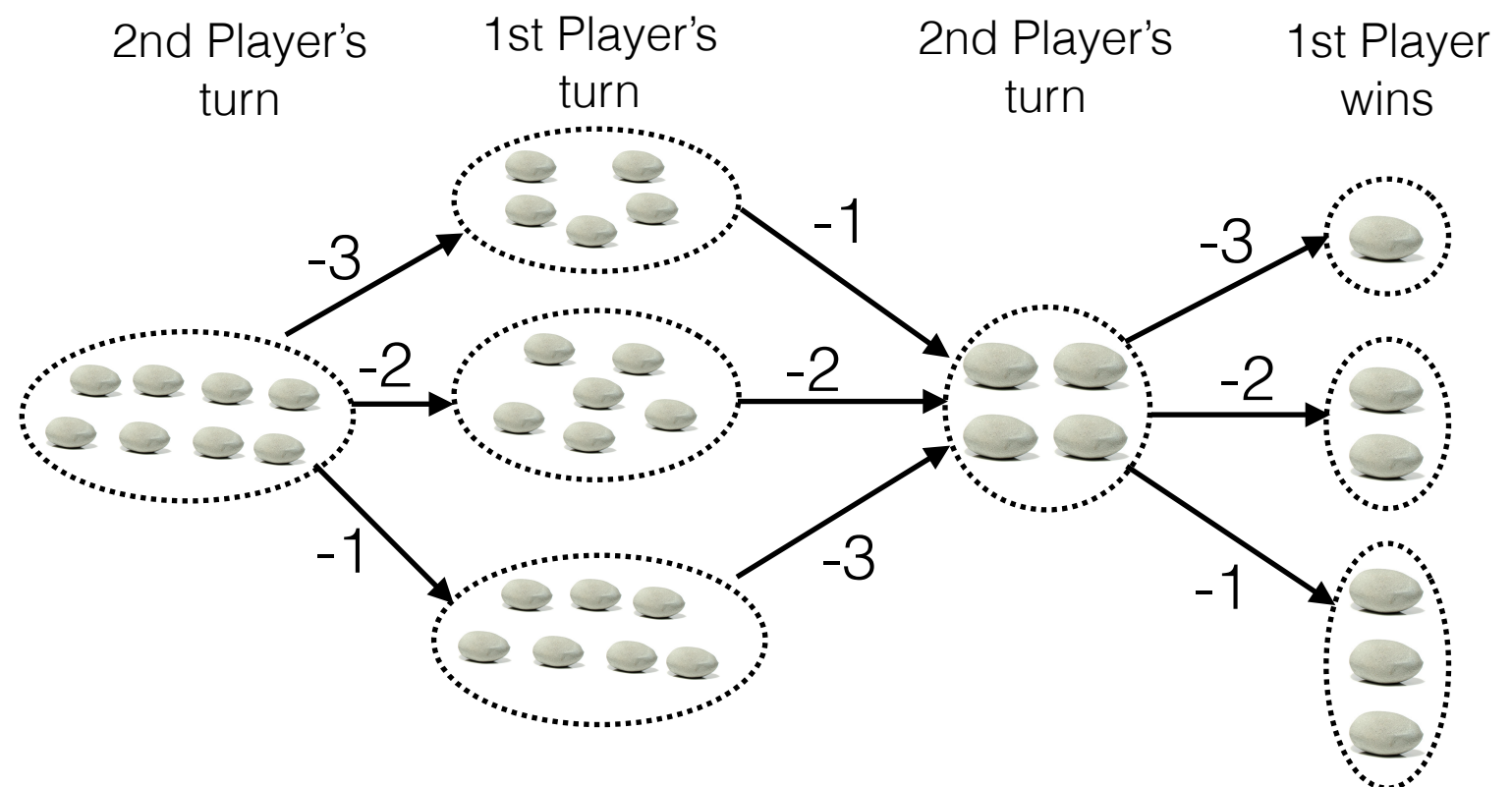


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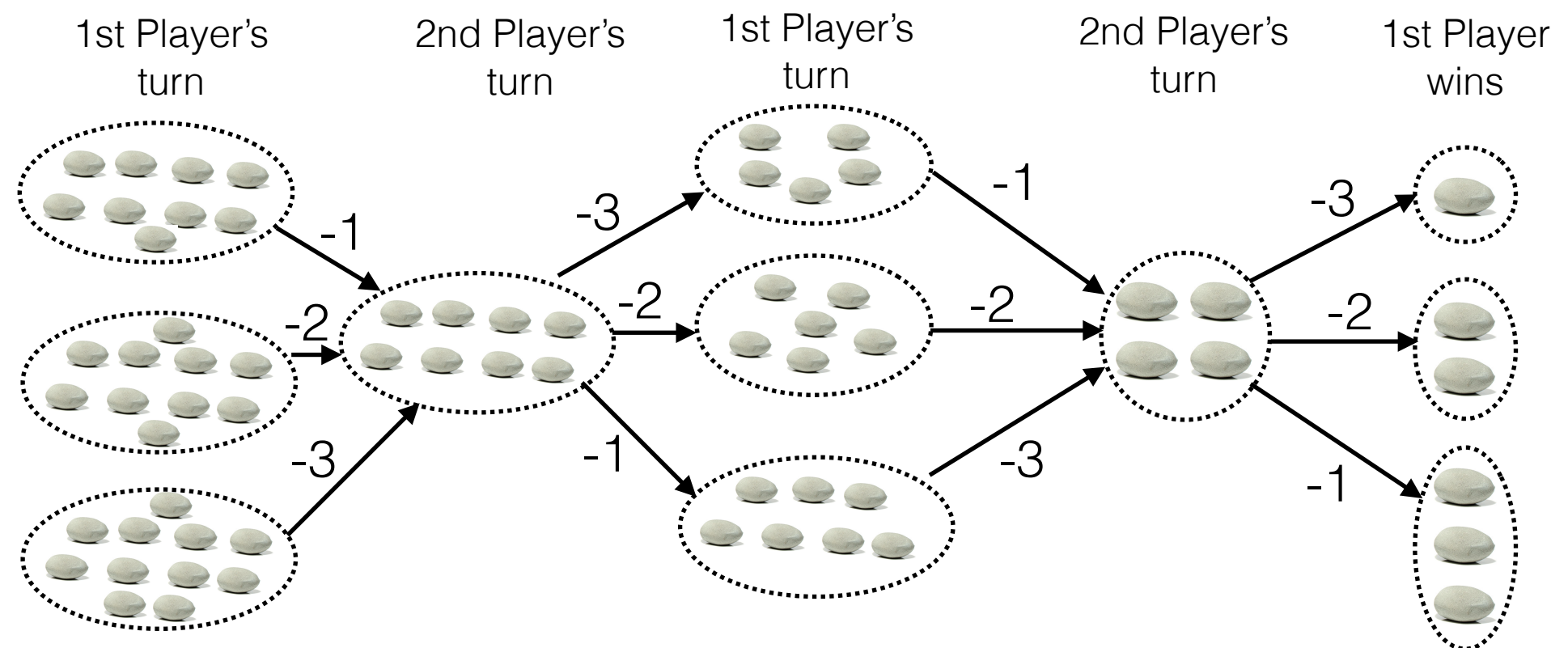


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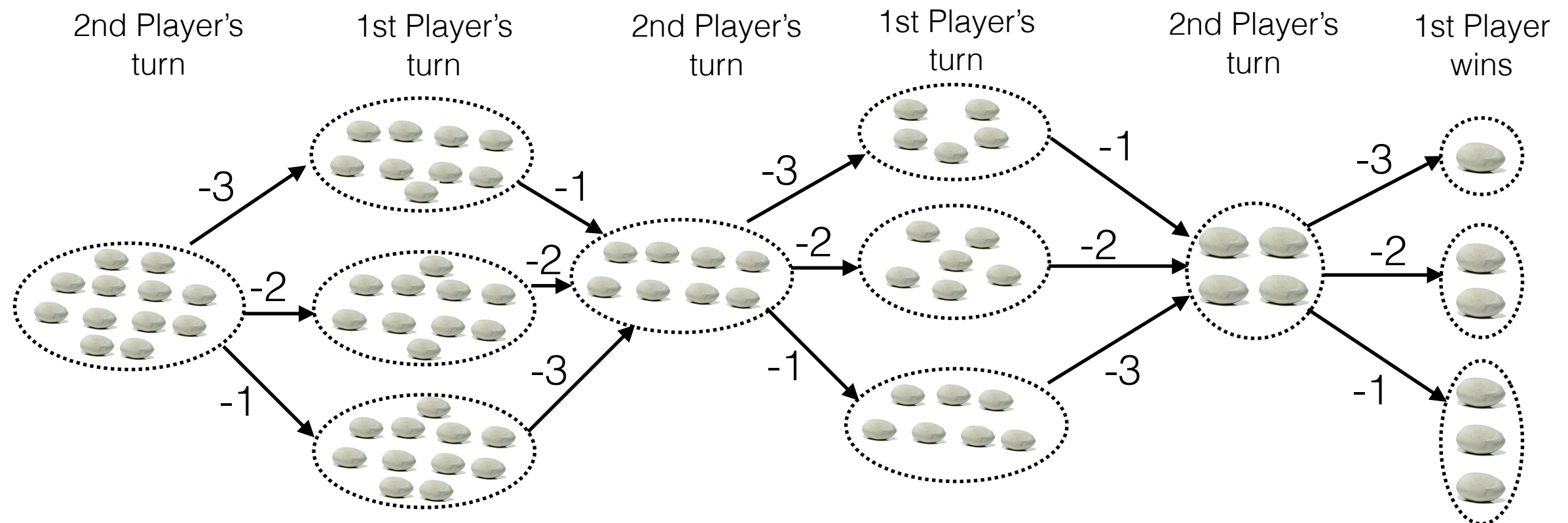


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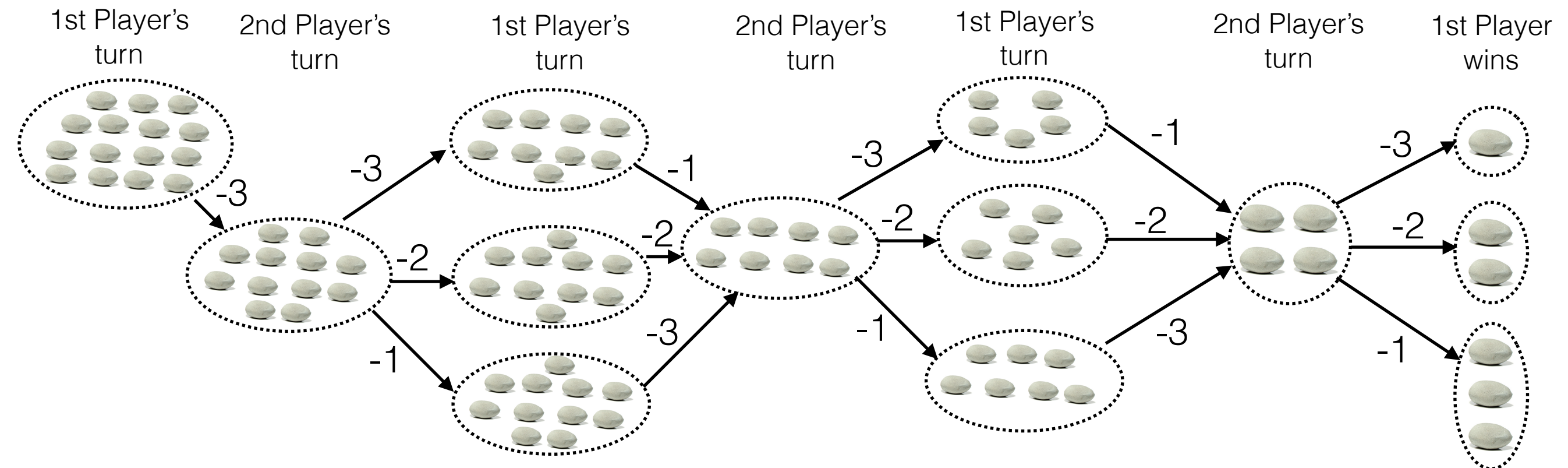


Forward and Backward Reasoning

Backward reasoning: Examples

Theorem. In a Nim game with 15 stones, the first player (the player who moves first) has a winning strategy.

Proof. Suppose the first player wins.



Adapting Existing Proofs

Experience

IS

the *Best* teacher

IN

LIFE

Adapting Existing Proofs

Experience

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IN

MATHEMATICS

Adapting Existing Proofs

Theorem. $\sqrt{3}$ is irrational.

Adapting Existing Proofs

Theorem. $\sqrt{3}$ is irrational.

Proof. Suppose $\sqrt{2}$ is rational.

Then $\sqrt{2} = m/n$ for some m and n s.t. $n \neq 0$ and $\gcd(m, n) = 1$.

Thus $2 = m^2/n^2$ and $m^2 = 2n^2$. So m^2 is even.

It follows that m is even, i.e. $m = 2k$ for some k .

We deduce that $m^2 = 4k^2 = 2n^2$ and $n^2 = 2k^2$. Thus n^2 is even.

It follows that n is even, i.e. $n = 2l$ for some integer l .

Therefore, $2 \mid \gcd(m, n)$, contradicting to $\gcd(m, n) = 1$.

Adapting Existing Proofs

Theorem. $\sqrt{3}$ is irrational.

Proof. Suppose $\sqrt{3}$ is rational.

Then $\sqrt{3} = m/n$ for some m and n s.t. $n \neq 0$ and $\gcd(m, n) = 1$.

Thus $3 = m^2/n^2$ and $m^2 = 3n^2$. So $m^2 \equiv 0 \pmod{3}$.

It follows that $m \equiv 0 \pmod{3}$, i.e. $m = 3k$ for some k .

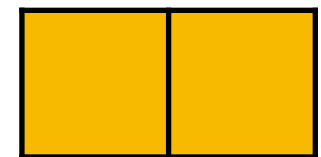
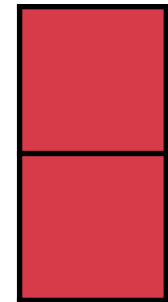
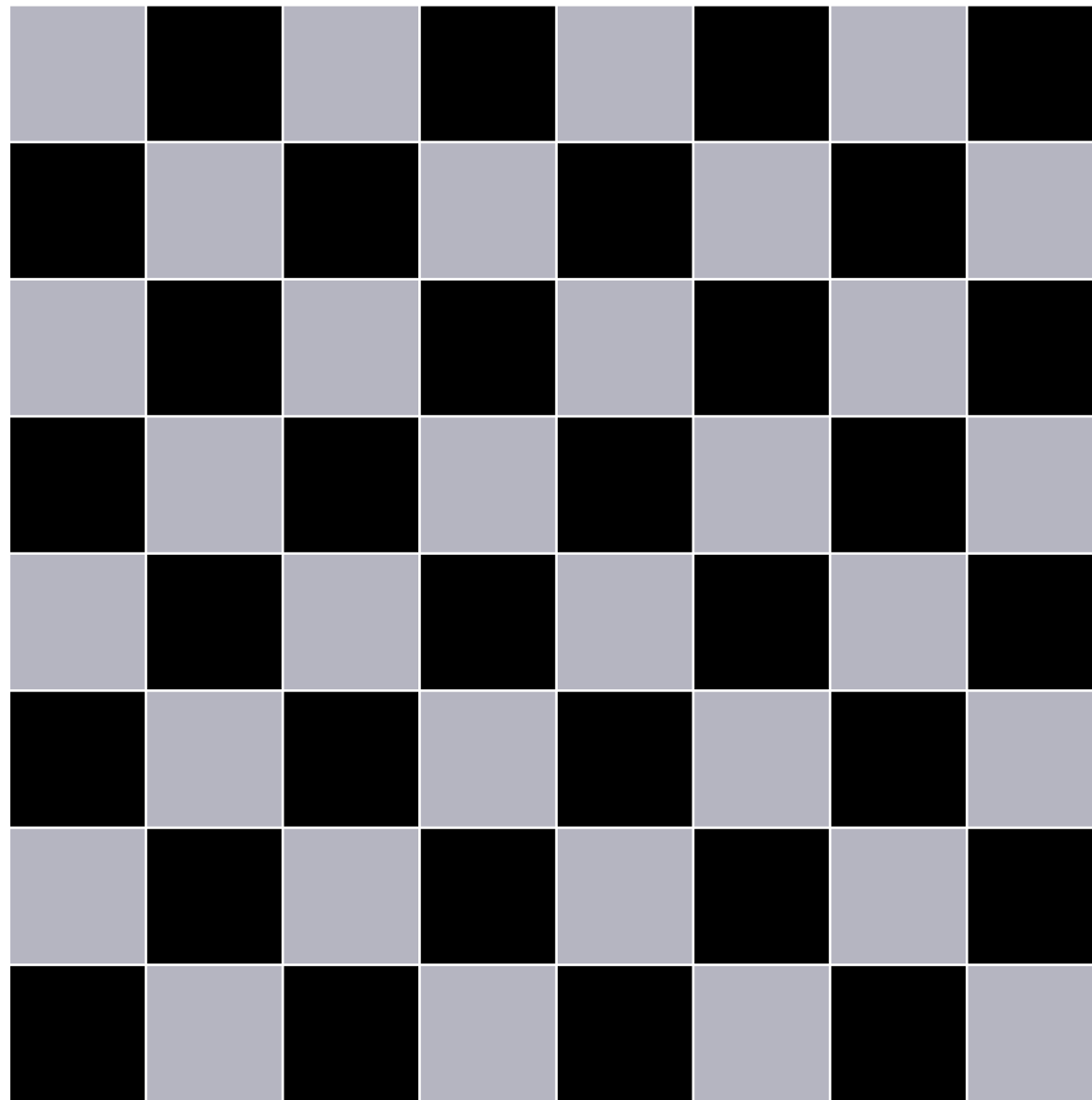
We deduce that $m^2 = 9k^2 = 3n^2$ and $n^2 = 3k^2$. Thus $n^2 \equiv 0 \pmod{3}$.

It follows that $n \equiv 0 \pmod{3}$, i.e. $n = 3l$ for some integer l .

Therefore, $3 \mid \gcd(m, n)$, contradicting to $\gcd(m, n) = 1$.

Illustration With Tilings

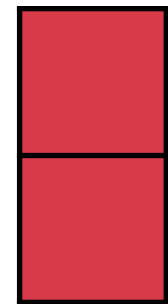
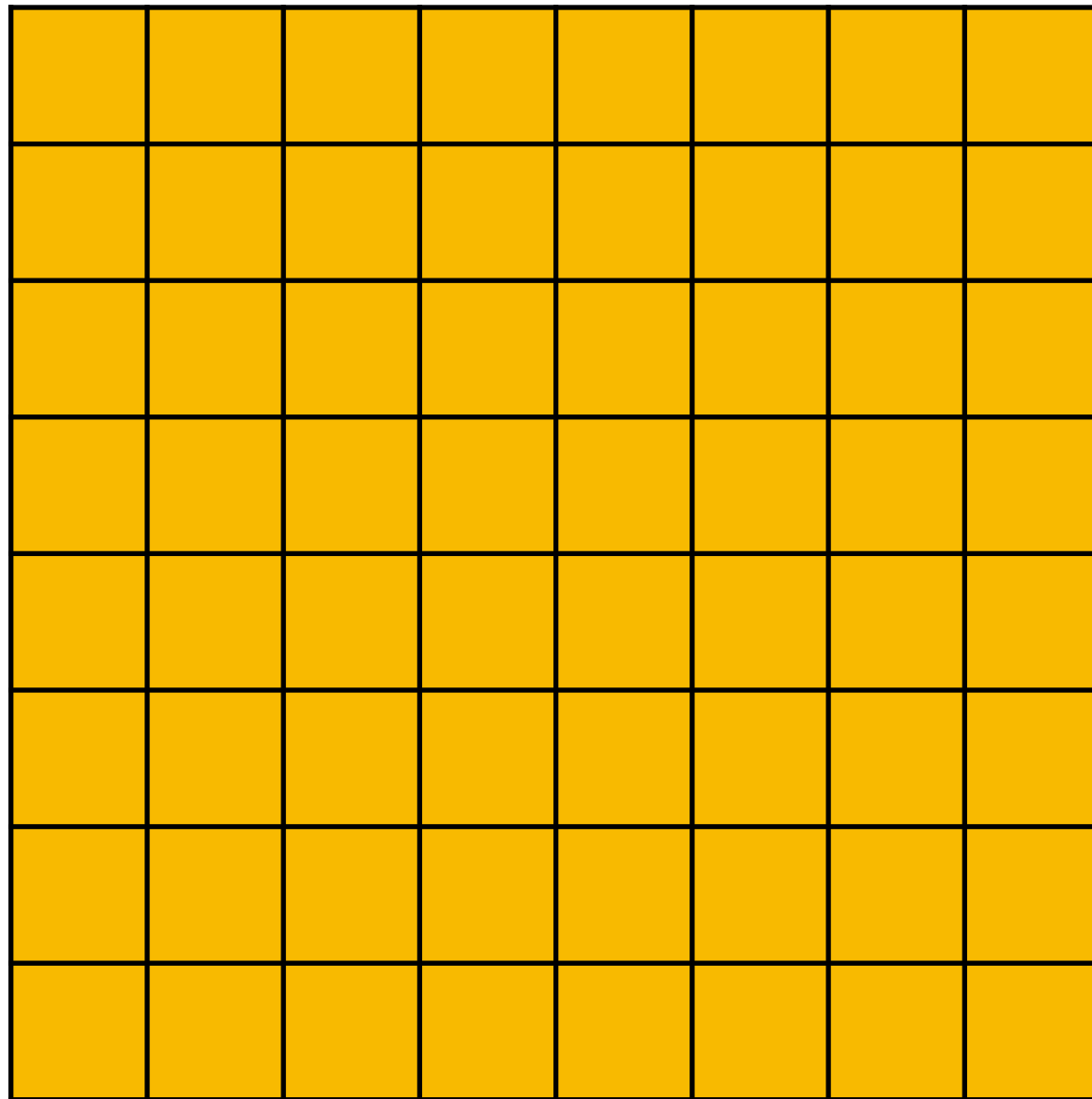
The standard checkerboard



Dominoes

Illustration With Tilings

A domino tiling

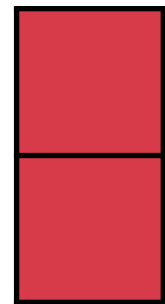
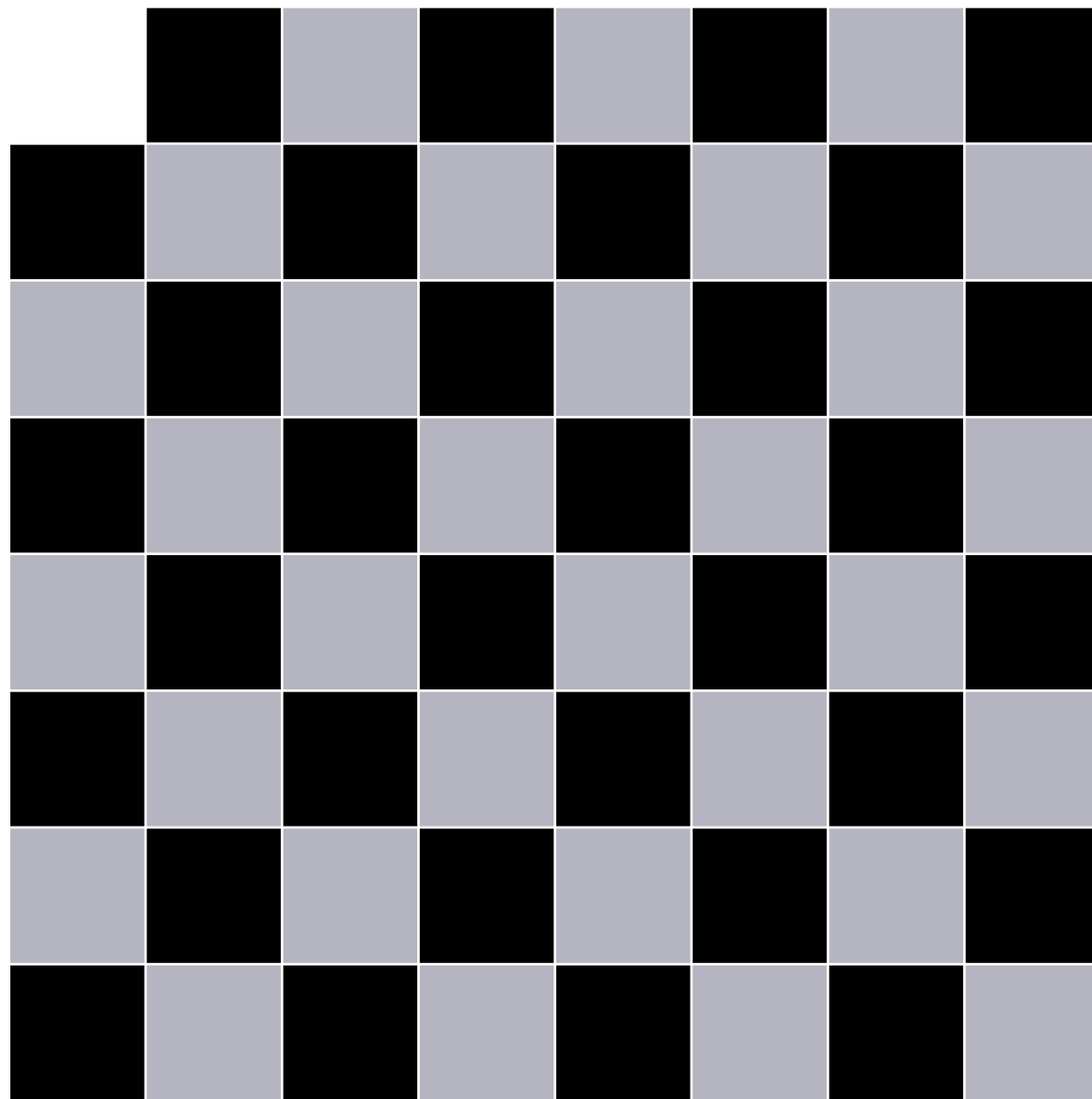


Dominoes

Non-overlapping dominoes cover the board exactly

Illustration With Tilings

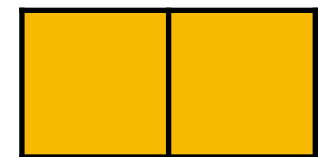
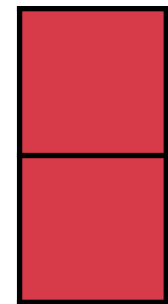
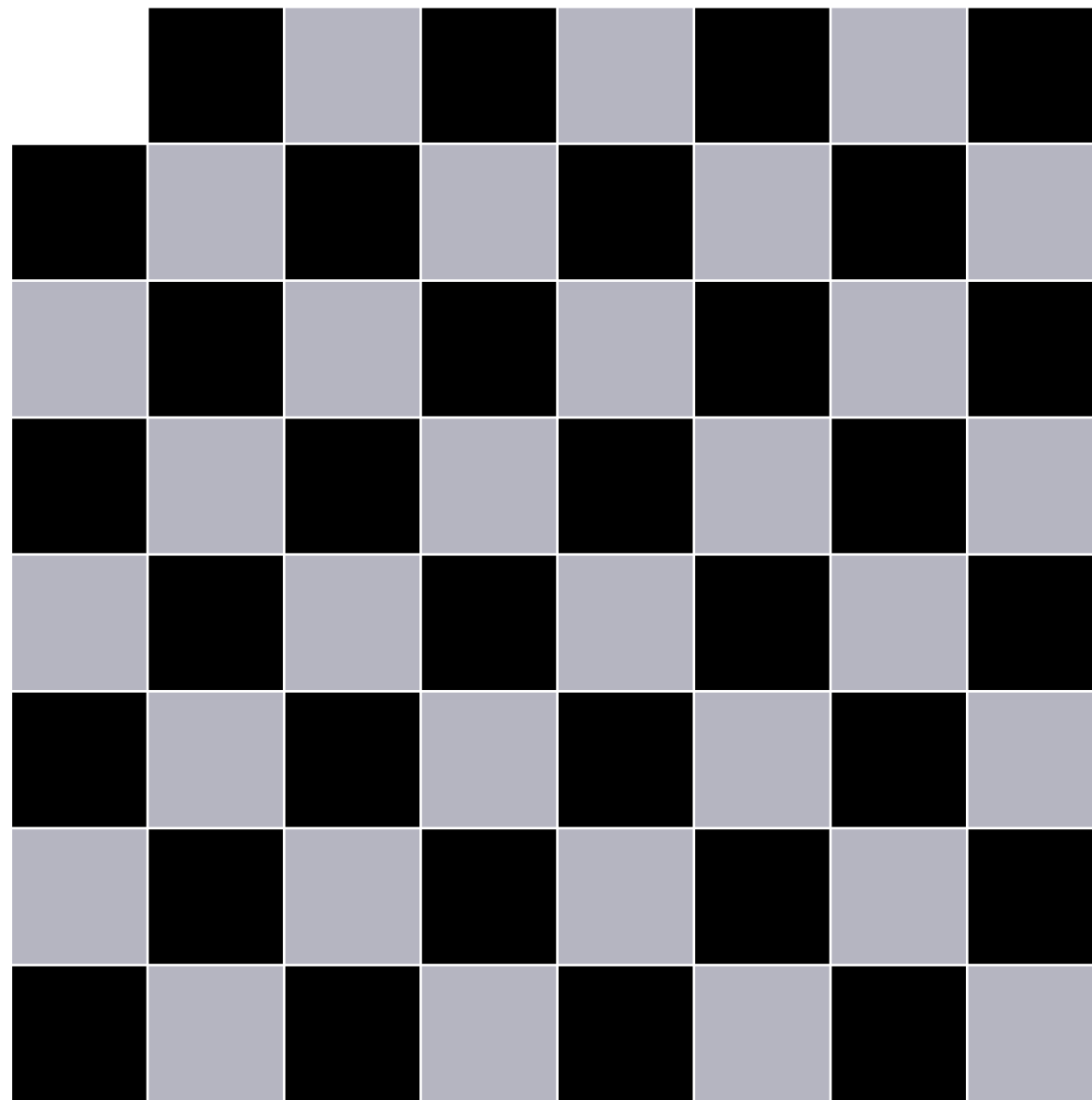
A domino tiling for this board ?



Dominoes

Illustration With Tilings

A domino tiling for this board ?

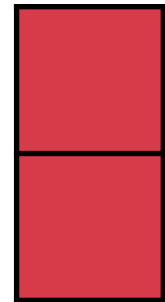
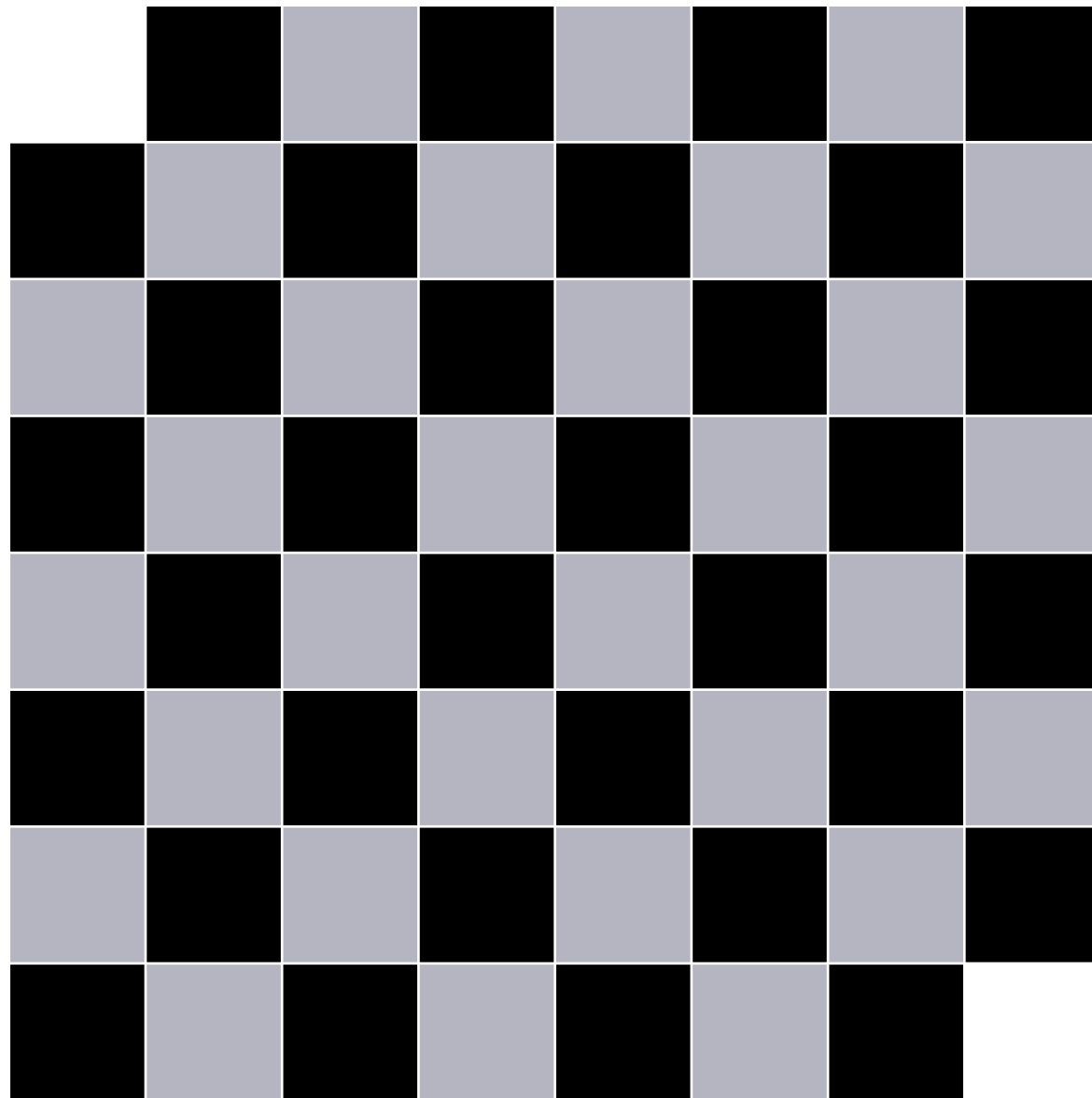


Dominoes

Impossible: board 63 cells, each domino covers 2 cells

Illustration With Tilings

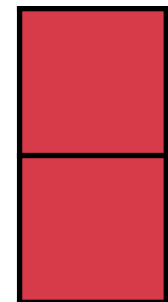
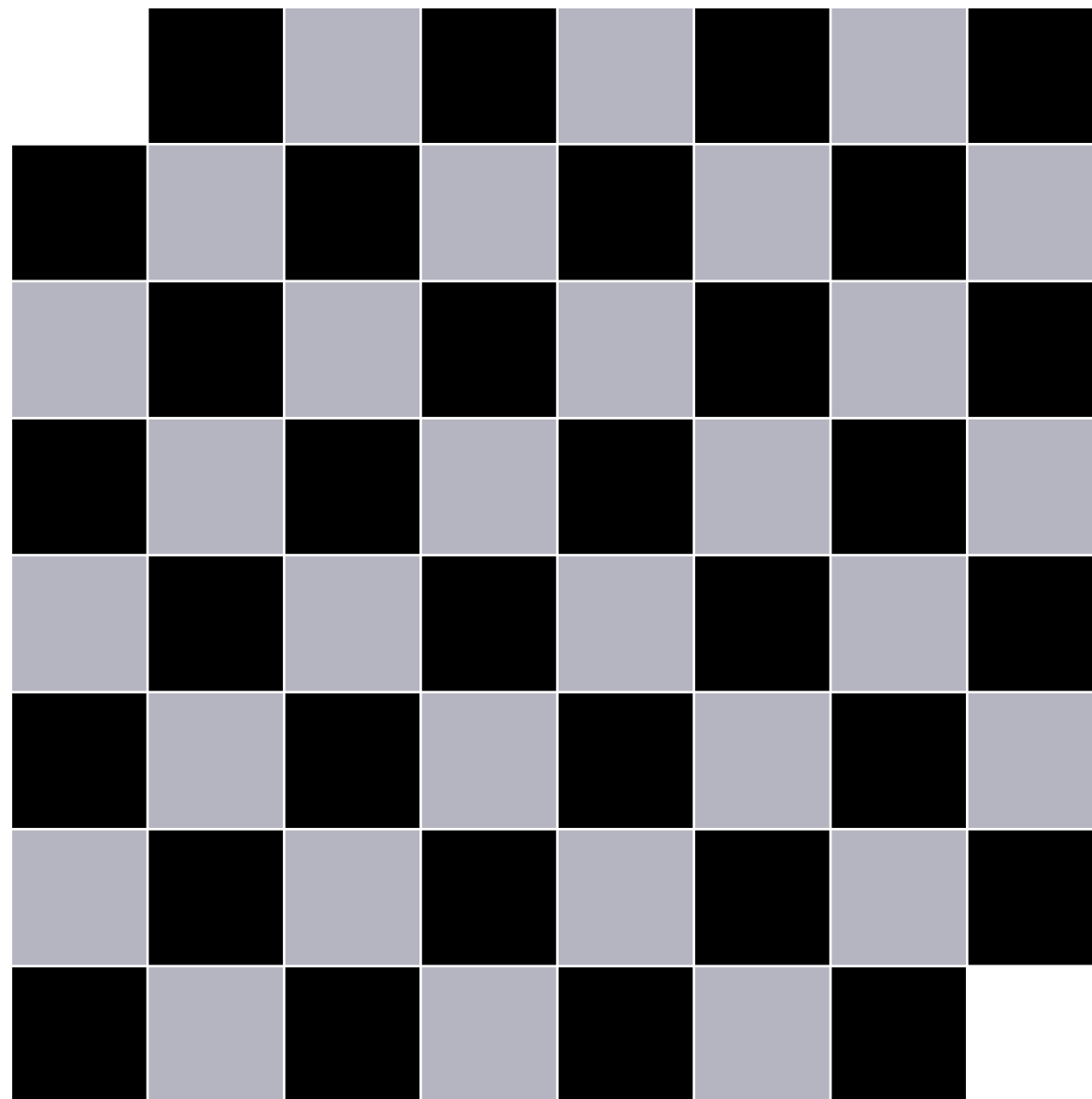
A domino tiling for this board ?



Dominoes

Illustration With Tilings

A domino tiling for this board ?

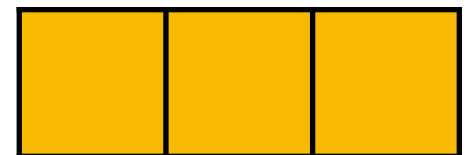
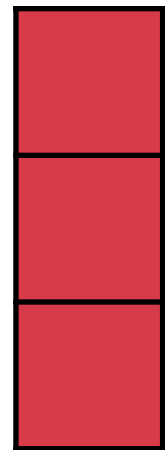
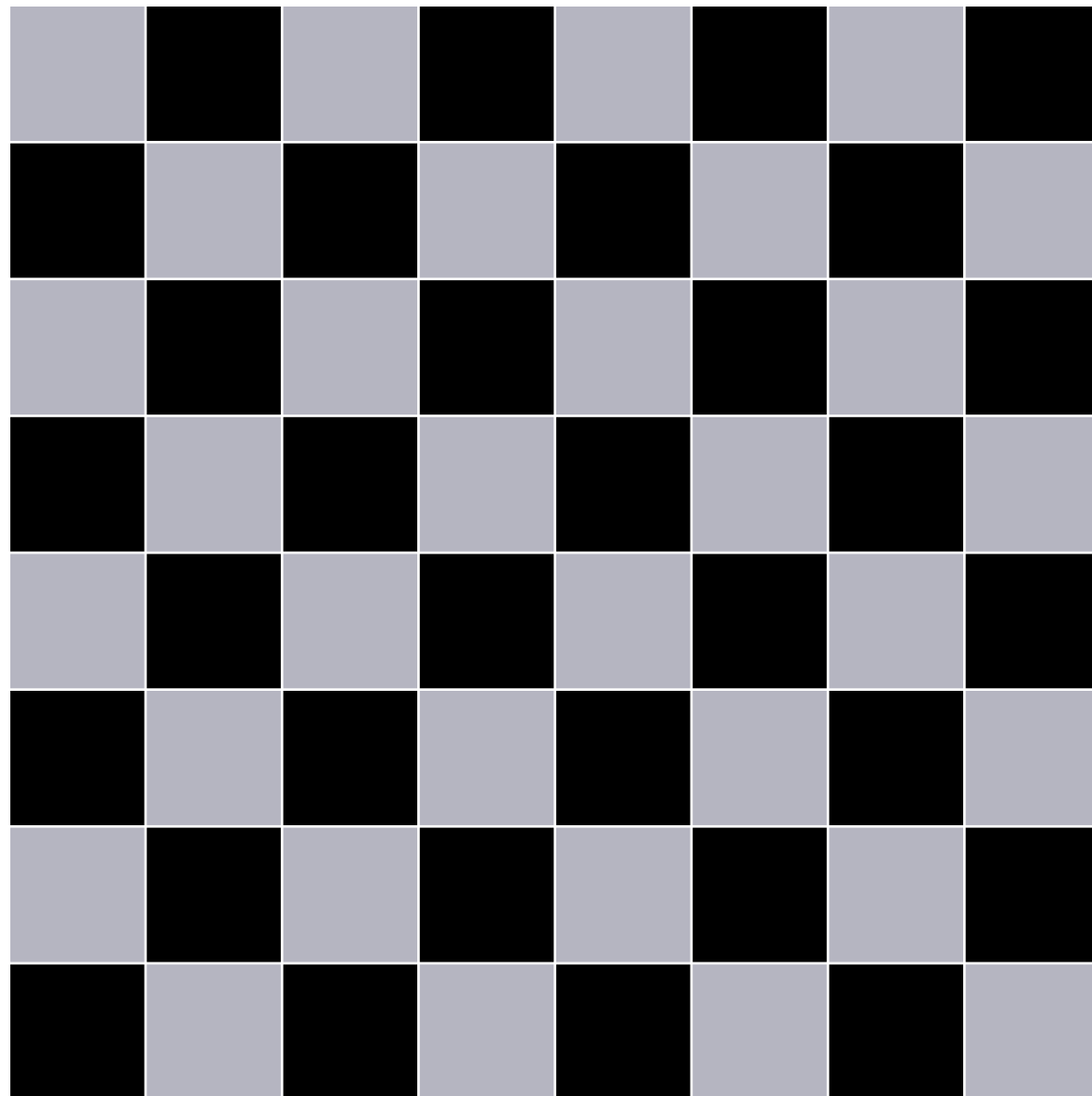


Dominoes

Impossible: board 32 black cells and 30 gray cells,
each domino covers 1 black and 1 gray cell.

Illustration With Tilings

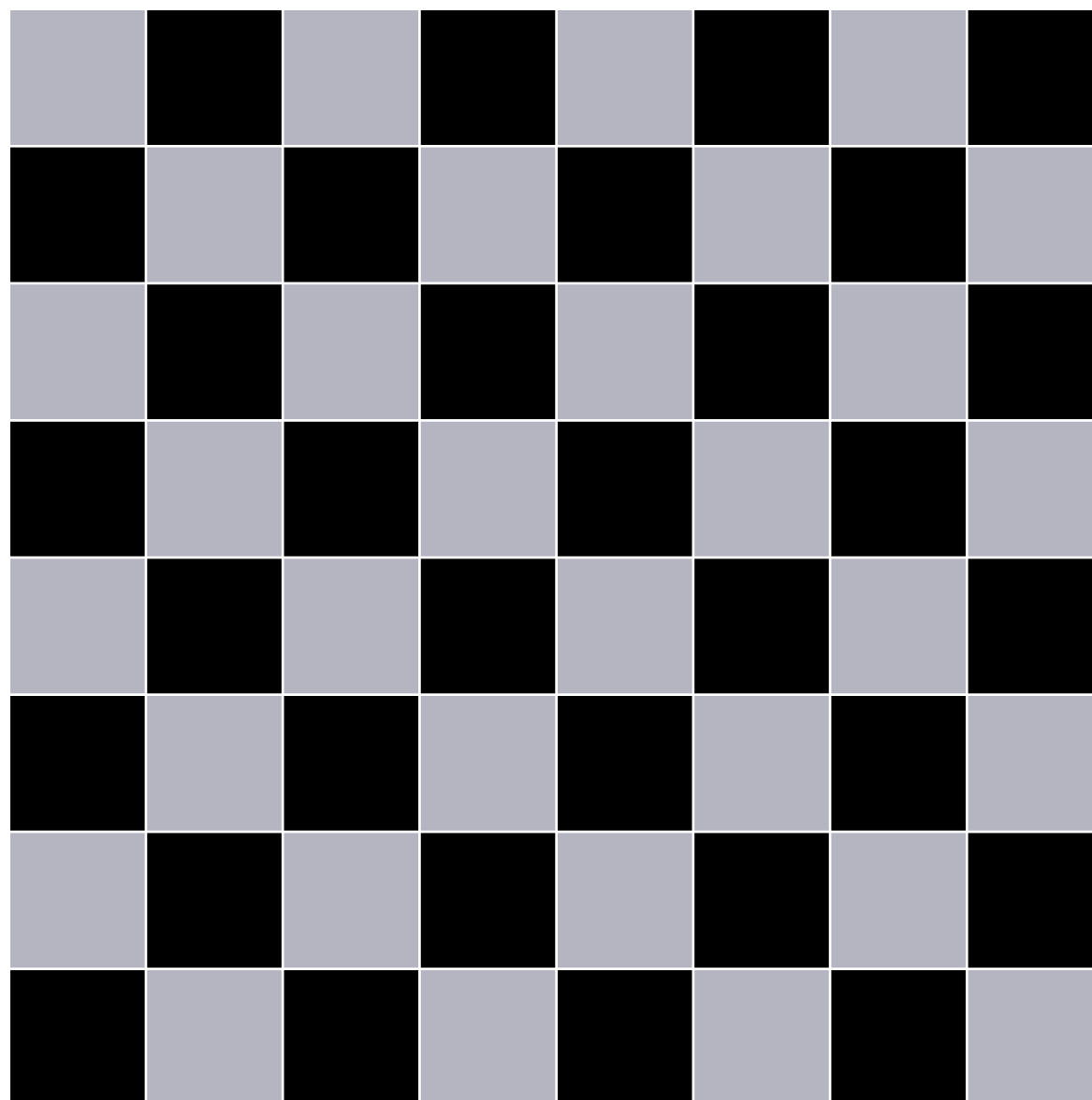
A triomino tiling for this board ?



Triominoes

Illustration With Tilings

A triomino tiling for this board ?

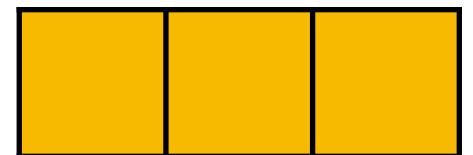
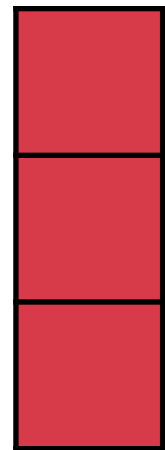
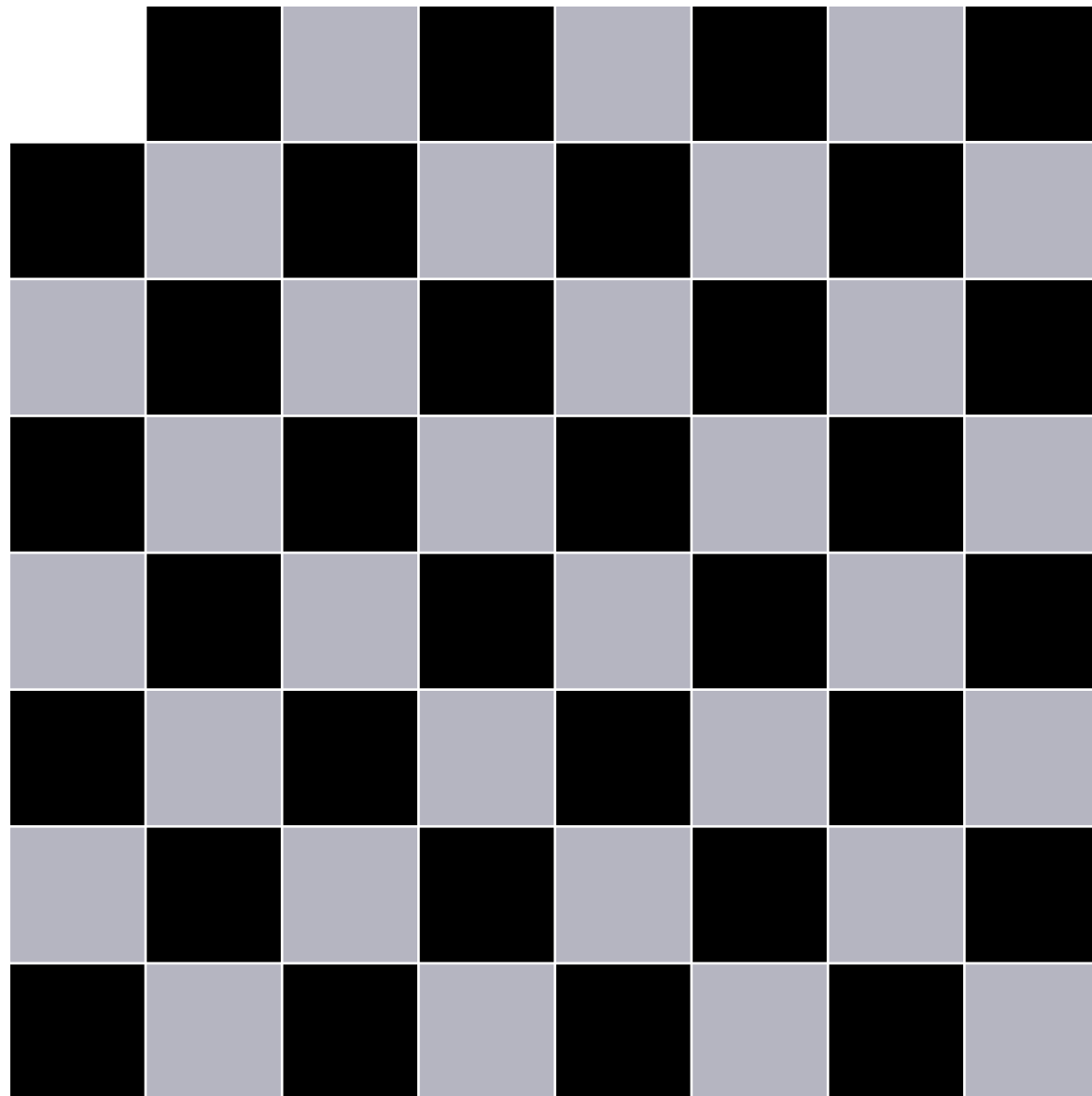


Triominoes

Impossible: board 64 cells,
each triomino covers 3 cells.

Illustration With Tilings

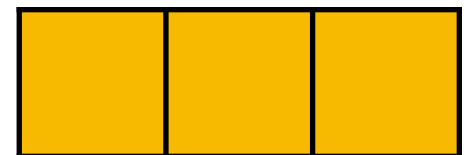
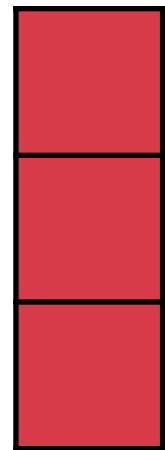
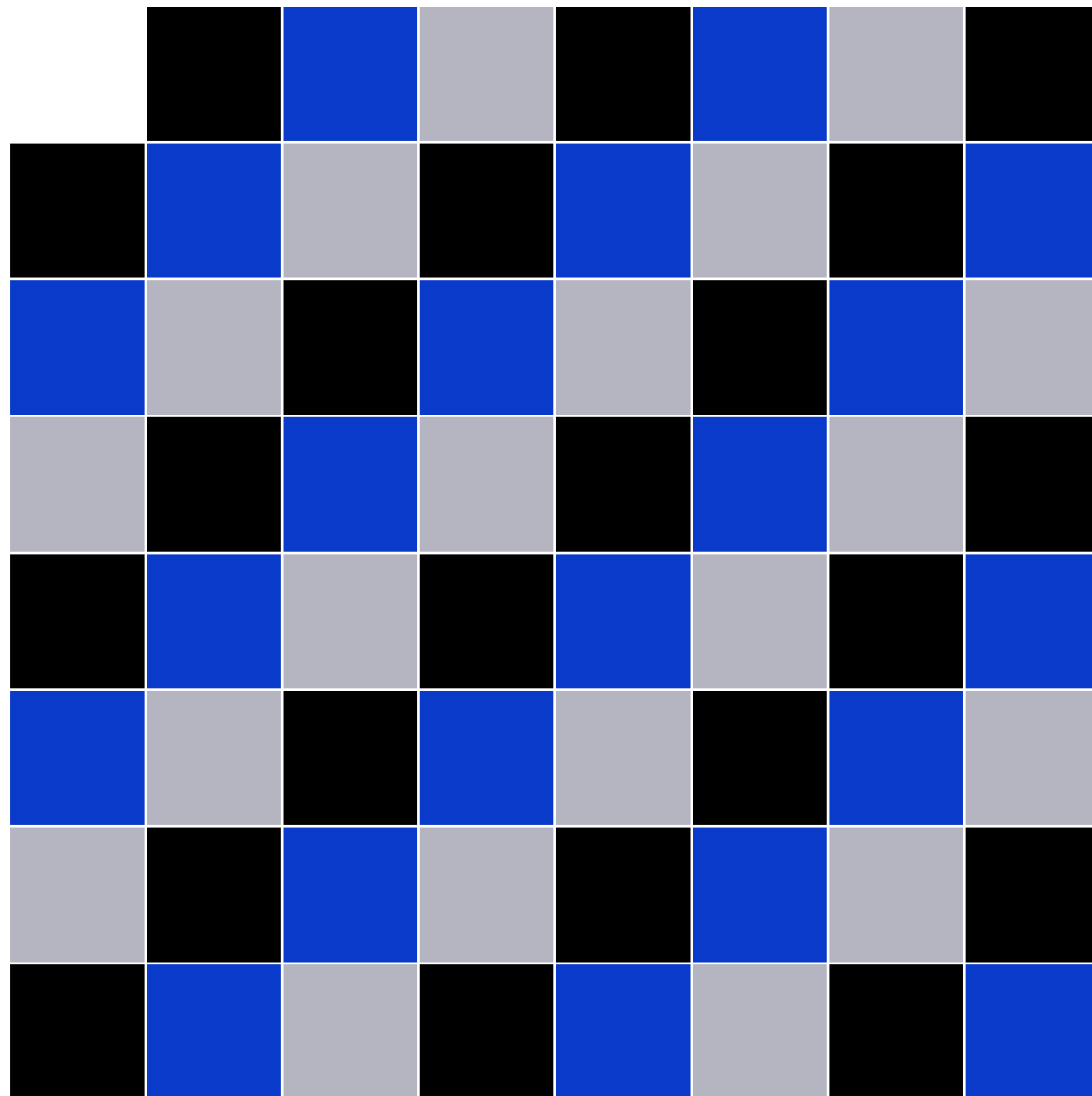
A triomino tiling for this board ?



Triominoes

Illustration With Tilings

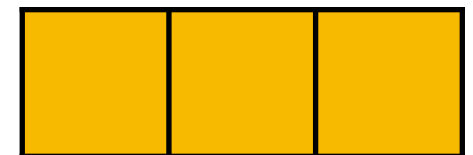
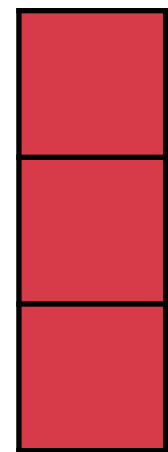
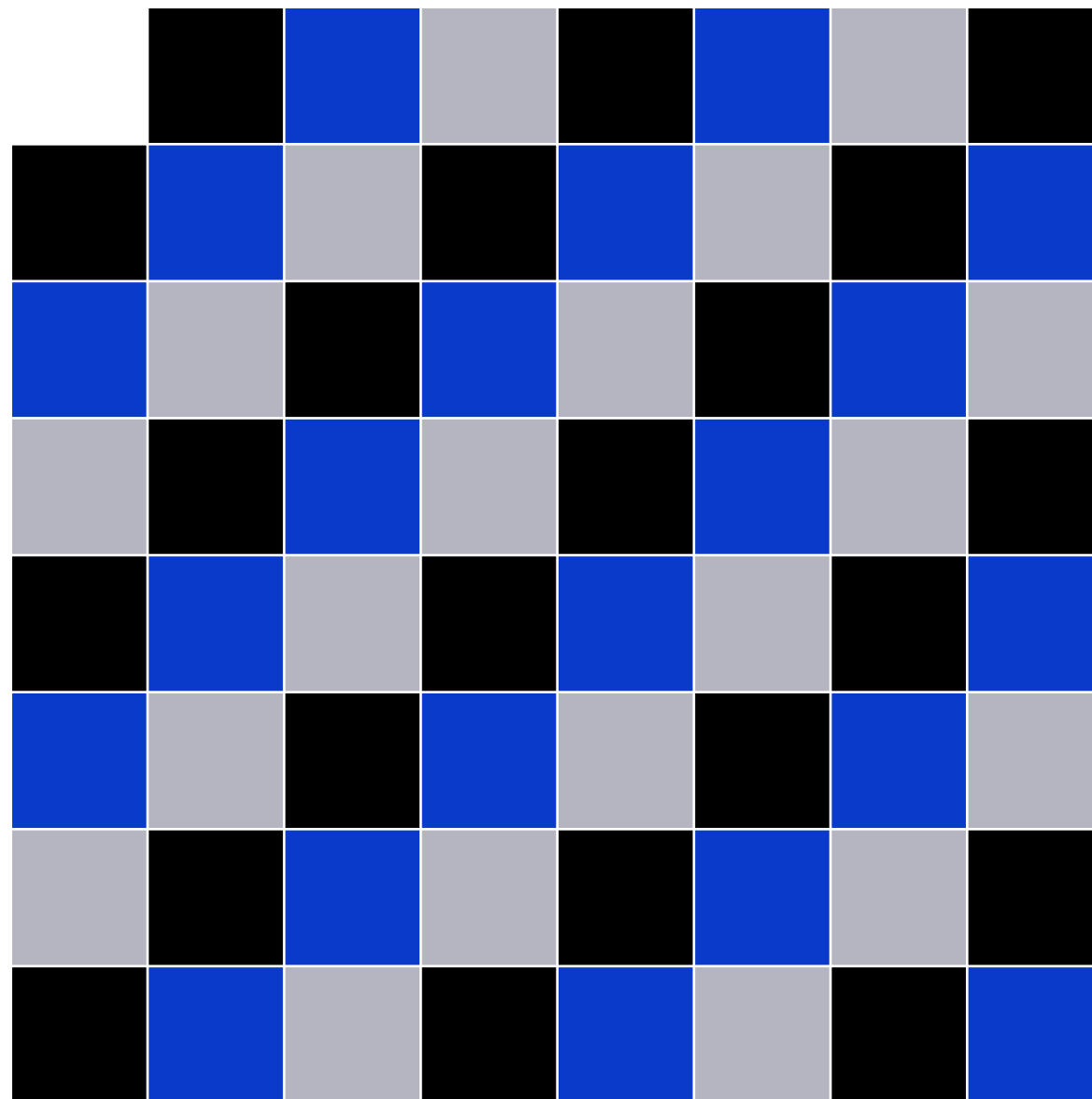
A triomino tiling for this board ?



Triominoes

Illustration With Tilings

A triomino tiling for this board ?



Triominoes

Impossible: board 20 gray cells, 21 blue cells, 22 black cells
each triomino covers 1 gray, 1 blue and 1 black cell.

Outline

- Proofs: Formal versus Informal
- Some terminology about proofs
- Proof methods
- Proof of equivalence
- Mistakes in proofs
- Proof strategies
 - Forward and backward reasoning
 - Adapting existing proofs
 - Illustration with tilings
- **The role of open problems**

The Role of Open Problems

Ferma's Last Theorem. The equation $x^n + y^n = z^n$ has no integer solutions s.t. $xyz \neq 0$ when $n > 2$.

The equation $x^2 + y^2 = z^2$ has infinitely many integer solutions, called Pythagorean triples, e.g. $3^2 + 4^2 = 5^2$.

Ferma's Last Theorem

- Claimed in the margin of a book to have a “wonderful proof” by Ferma in 17th century.
- Was essentially a conjecture for more than 300 years.
- $n = 3$ proved by Euler, $n = 4$ proved by Ferma himself.
- Eventually proved by Andrew Wiles (Oxford) in 1994.

The Role of Open Problems

$3x+1$ conjecture (Collatz problem).

Let $f(x) = \begin{cases} x/2 & \text{if } x \equiv 0 \pmod{2} \\ 3x + 1 & \text{otherwise} \end{cases}$. Then for any $x \geq 1$, if we repeatedly apply f , then eventually we will reach 1.

$7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow \dots \rightarrow 2 \rightarrow 1$

Story about the $3x+1$ conjecture

- Has been verified for 5.6×10^{13} .
- Attracted the attention of many mathematicians since the 1950s.
- Raised many times and has many names, e.g. Hasse's algorithm, Ulam's problem, the Syracuse problem, and Kakutani's problem.
- Joked as a conspiracy to slow down the American math. research.

Recap

● Proof methods

Theorem. $\forall x (\varphi_1(x) \rightarrow \varphi_2(x))$.

Direct proof
 $\varphi_1(c)$ implies $\varphi_2(c)$

Proof by contraposition
 $\neg\varphi_2(c)$ implies $\neg\varphi_1(c)$

Proof by contradiction
 $\varphi_1(c) \wedge \neg\varphi_2(c)$ implies F

Proof by cases

If $\varphi_1 \equiv \varphi_{1,1} \vee \dots \vee \varphi_{1,n}$, then show $\varphi_{1,1}(c)$ implies $\varphi_2(c)$, $\dots$, and $\varphi_{1,n}(c)$ implies $\varphi_2(c)$

Existence proof: constructive and
 nonconstructive proof

1729 = $10^3 + 9^3 = 12^3 + 1^3$ xy is a rational for either
 Uniqueness proof $x=y=\sqrt{2}$ or $(x=\sqrt{2}^{\sqrt{2}} \text{ and } y=\sqrt{2})$

● Proof of equivalence

Theorem. $\forall x \varphi_1(x) \leftrightarrow \varphi_2(x) \dots \leftrightarrow \varphi_n(x)$.

Show $\forall x \varphi_1(x) \rightarrow \varphi_2(x)$, $\dots$, and $\forall x \varphi_n(x) \rightarrow \varphi_1(x)$.

● Mistakes in proofs

In the proof, assume
 both φ and $\neg\varphi$

$$\frac{\varphi_2}{\varphi_1 \rightarrow \varphi_2} \quad \therefore \varphi_1$$

$$\frac{\neg\varphi_1}{\varphi_1 \rightarrow \varphi_2} \quad \therefore \neg\varphi_2$$

Circular reasoning
 $p \leftrightarrow q$ does not imply p and q

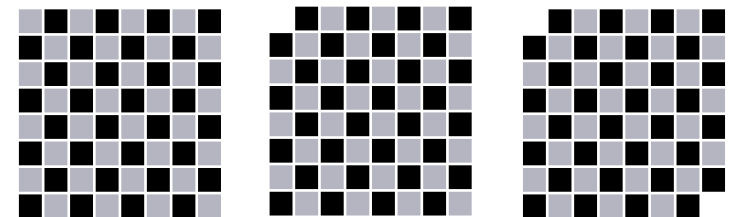
Recap (Continued)

- Proof Strategies



The first player always wins in the Nim game with 15 stones

- Forward and backward reasoning
- Adapting existing proofs $\sqrt{3}$ is irrational
- Illustration with tilings



Domino



triomino



- The role of open problems

$$x^n + y^n = z^n \quad f(x) = \begin{cases} x/2 & \text{if } x \equiv 0 \pmod{2} \\ 3x + 1 & \text{otherwise} \end{cases}$$

Homework

1. Section 1.7

- Page 96, Exercise 25, 34, 42

2. Section 1.8

- Page 113-114, Exercise 12, 18, 42

作业和答疑

1. 交作业时间：奇数周周一上课前（两周交一次）
2. 交作业方式（二选一）：
 - ◆ 通过<https://mooc.ucas.edu.cn/portal>课程网站
 - ◆ 上课现场提交
3. 答疑：助教崔乐天、刘易铖

Next Lecture

Divisibility, Modular Arithmetic,
Representations of Integers